STUDENT PERFORMANCE ANALYSIS

End-to-End Data Engineering & Business Intelligence Project

Project brief and implementation assignment

Database	PostgreSQL
Database client	DBeaver
Visualization	Microsoft Power BI
Source data	Student-level CSV dataset

From raw CSV data to a relational database, analytical SQL, and an interactive Power BI dashboard.

1. Project Overview

You have been provided with a CSV dataset containing student information, academic performance, financial circumstances, learning activities, attendance, and personal well-being indicators. Your responsibility is to design and implement a complete data management solution using PostgreSQL and DBeaver. You will ingest the data, design the database structure, populate the database tables, and perform analytical queries to identify meaningful insights about student performance.

After completing the database and SQL analysis, you will use Microsoft Power BI to develop an interactive dashboard and present your findings.

2. Project Objectives

- Understand database design principles.
- Differentiate between Conceptual Data Modelling (CDM) and Logical Data Modelling (LDM).
- Understand database normalization and relationships.
- Create staging tables in PostgreSQL and import CSV data using DBeaver.
- Design database tables with appropriate primary keys and foreign keys.
- Distribute data from a staging table into multiple relational tables.
- Write SQL queries to extract meaningful insights.
- Explore relationships between academic performance and student-related factors.
- Build interactive dashboards using Power BI and present evidence-based findings.

3. Tools and Technologies

Tool	Purpose
PostgreSQL	Relational database management system
DBeaver	Database development and management
Microsoft Power BI	Data visualization and reporting
SQL	Data transformation and analytical queries

4. Dataset Description

The provided CSV file contains student-level information across the following categories:

A. Student Demographics: person_id, age, gender, nationality, first_generation, disability_status, international_student

B. Financial Background: family_income_level, scholarship_holder, tuition_up_to_date, part_time_job, weekly_work_hours

C. Academic Information: course_type, study_mode, semester, admission_grade, previous_qualification_grade, units_enrolled, units_passed, units_failed, gpa_semester_1, gpa_semester_2

D. Learning Engagement: attendance_rate, assignments_submitted_rate, library_visits_per_month, lms_logins_per_week, extracurricular_activities, missed_deadlines

E. Student Well-being: stress_level, mental_health_support, satisfaction_score, commute_time_mins, sleep_hours

F. Academic Outcome: dropout

5. Project Implementation

Phase 1: Environment Setup

- Install PostgreSQL.
- Install DBeaver.
- Configure a PostgreSQL database connection in DBeaver.
- Create a new database named `student_performance_db`.

Phase 2: Data Ingestion – Staging Table

- Examine the provided CSV file and understand its structure.
- Identify the data types of all columns.
- Create a staging table in PostgreSQL containing all columns available in the CSV.
- Import the complete CSV dataset into the staging table using DBeaver.
- Verify that the imported record count matches the CSV record count.
- Check for missing values, duplicate records, and inconsistent data.

Expected output: A populated staging table containing the complete source dataset.

Phase 3: Database Modelling

A. Conceptual Data Model (CDM)

- Create a high-level representation of the database.
- Identify the major entities and the business concepts represented by the dataset.
- Define the relationships between entities.
- Show how the entities relate to one another.

B. Logical Data Model (LDM)

- Convert the conceptual model into a detailed logical structure.
- Define all entities and their attributes.
- Identify primary keys and foreign keys.
- Define relationships and cardinality.
- Apply appropriate normalization principles and avoid unnecessary data redundancy.

Expected output: A documented CDM and LDM diagram.

Phase 4: Database Implementation

- Create the required database tables in PostgreSQL based on your CDM and LDM.
- Define appropriate data types for each column.
- Implement primary key and foreign key constraints.
- Define relationships between the tables.
- Apply normalization principles wherever appropriate.
- Ensure referential integrity across the tables.

Phase 5: Data Transformation and Loading

- Extract the relevant columns from the staging table.
- Transform the data according to the target table structures.
- Remove duplicate records where necessary.
- Handle missing and inconsistent values appropriately.

- Insert the transformed data into the respective tables.
- Ensure that foreign key relationships are maintained.
- Validate final record counts and data consistency.

Expected output: All final database tables populated with the appropriate data from the staging table.

6. SQL Analysis and Business Questions

After populating the database, answer the following 15 questions using SQL. For every question, provide the SQL query, the result, and a brief interpretation of the finding.

Easy Level – Basic Data Analysis

1. What is the total number of students in the dataset?

2. What is the distribution of students across different genders?

3. How many students belong to each nationality?

4. What is the average GPA for Semester 1 and Semester 2?

5. What is the total number of students who have dropped out versus those who have continued their studies?

Hard Level – Comparative Analysis

6. What is the dropout rate across different course types?

7. How does the average GPA differ between scholarship holders and non-scholarship holders?

8. Is there a relationship between attendance rate and the number of units passed? Compare students with different attendance levels.

9. How does working part-time affect academic performance? Compare the average GPA of students who work part-time against those who do not, considering their weekly working hours.

10. Which factors are associated with higher dropout rates: family income level, tuition payment status, or scholarship status? Compare dropout rates across these categories.

Advanced Level – Multidimensional Analysis

11. Does student engagement influence academic performance and dropout rates? Analyze the combined relationship between attendance rate, LMS logins, library visits, assignment submission rate, and missed deadlines.

12. How does student well-being relate to academic performance? Analyze the relationship between stress level, sleep hours, mental health support, satisfaction score, and GPA.

13. Are first-generation students more likely to drop out than non-first-generation students? Compare dropout rates while also considering scholarship status and family income level.

14. Does commuting time influence student attendance, academic performance, and dropout rates? Identify any patterns across different commuting-time groups.

15. Identify the top five student-related factors associated with dropout. Support your findings with appropriate SQL analysis, comparisons, and statistical summaries. Clearly distinguish between correlation and causation.

Analytical expectation: Questions 11–15 should go beyond simple GROUP BY queries. Consider using CTEs, CASE expressions, window functions, conditional aggregation, and statistical comparisons where appropriate.

7. Power BI Dashboard Development

After completing the SQL analysis, connect Power BI to your PostgreSQL database and develop an interactive dashboard.

A. Student Overview

- Total students
- Gender distribution
- Nationality distribution
- International versus domestic students
- First-generation student distribution

B. Academic Performance

- Average GPA by semester
- Units passed versus failed
- Performance by course type
- Attendance and assignment submission trends

C. Financial Analysis

- Scholarship holder distribution
- Family income distribution
- Tuition payment status
- Academic performance by financial category

D. Student Engagement and Well-being

- LMS logins
- Library visits
- Extracurricular participation
- Stress levels
- Sleep hours
- Satisfaction scores

E. Dropout Analysis

- Total dropout count
- Overall dropout rate
- Dropout rate by course type
- Dropout rate by financial category
- Dropout rate by attendance and engagement levels
- Key factors associated with student dropout

Include appropriate slicers and filters to make the dashboard interactive.

8. Findings and Presentation

Prepare a final presentation covering:

- Project overview and objectives.
- Dataset description.
- Conceptual Data Model.
- Logical Data Model.
- Database architecture and relationships.
- Data ingestion and transformation process.
- SQL queries and analytical findings.
- Power BI dashboard.
- Key observations and recommendations.

Your findings should be supported by actual results from the dataset.

9. Final Deliverables

#	Deliverable
1	PostgreSQL database
2	Staging table with imported CSV data
3	Conceptual Data Model (CDM)
4	Logical Data Model (LDM)
5	SQL scripts for table creation
6	SQL scripts for data transformation and insertion
7	SQL scripts answering all 15 questions
8	Power BI dashboard (.pbix)
9	Final presentation with findings and recommendations

10. Evaluation Criteria

- Correct installation and configuration of tools.
- Proper database modelling and design.
- Correct implementation of primary and foreign keys.
- Appropriate normalization and data organization.
- Accurate data ingestion and transformation.
- SQL query correctness and analytical depth.
- Dashboard design and visualization quality.
- Ability to explain findings and draw meaningful conclusions.

Important: Do not simply generate SQL queries and dashboards without understanding the underlying concepts. You should be able to explain why you designed the database in a particular way, how data moves between tables, and what each analytical finding tells us about student performance.

Data modelling note: The `person_id` field may represent one record per student, but verify this in the source data. If students have multiple semester records, student information and academic performance may need to be modelled separately.